

Sadhana Jamkatel

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EDUCATION

North Central College, Naperville, IL **GPA 3.85/4** **Expected: May 2027**

Major: Bachelor of Science in Computer Science, **Minor:** Economics, **Concentration:** Ethical Leadership

Relevant Coursework: Python, Data Structures, Discrete Structures, Computer Organization and Design, Software Development, Website Development, Data Mining, Calculus I & II, Database System, Algorithm & Design, Computer Network & Security

TECHNICAL SKILLS

Programming Languages: Python, C/C++, JavaScript, TypeScript, Java

Web Development: React.js, Next.js, Node.js, Flask, HTML, CSS, Bootstrap

Cloud & Tools: AWS (S3, EC2, Lambda, CloudFront, CloudWatch, IAM, CDK), Docker, Linux/Unix, GitHub Actions, GitLab, Git, OCLIF

Databases: MongoDB, PostgreSQL, MySQL, Google Cloud Firestore

APIs & Frameworks: RESTful APIs, Flask, Microservices architecture

ML/AI: scikit-learn, pandas, NumPy, LangChain, Gemini, HuggingFace, CUDA

Other: Agile, CI/CD, Testing (pytest, Jest), Automation, OOP, Figma, Tableau, Cybersecurity, JSON/YAML

WORK EXPERIENCE

DevOps Engineer Intern| Intern of the Season 2025

June 2025-Aug 2025

Aeropay - Chicago, IL

- **Reduced documentation search time from 30 minutes to 30 seconds** by building a full-stack internal documentation site using **Docusaurus** (MDX, custom charts, integrated search), serving 30+ engineers daily and hosted on **GitHub Pages** with automated **CI/CD** via **GitHub Actions**
- Secured and deployed the site using **AWS S3 + CloudFront** with **JumpCloud SSO** access controls and **AWS CloudWatch** telemetry for performance monitoring, ensuring 99.9% uptime for DevOps and engineering teams
- **Reduced developer setup time by ~70%** by designing a **CLI tool in TypeScript** that automated authentication, role-based access, and token generation for AWS/database connections
- Automated CLI release documentation using **OCLIF** and **GitHub Actions**, publishing updated docs to a live site with every release and eliminating manual updates

High-Performance Computing (HPC) Bootcamp Participant

August 2025

Argonne National Laboratory - IL

- Built an interactive HPC chatbot for scientific Q&A; delivered RAG and non-RAG versions (LangChain , Huggingface, Gemini), compared accuracy and latency, and documented trade-offs.
- Gained hands-on exposure to Argonne's supercomputing ecosystem (incl. Aurora) and HPC workflows used in scientific computing.
- Completed workshops in data visualization and machine learning for science; practiced turning datasets into clear, usable insights.

PROJECTS

Predictive Maintenance ML System for Fleet Engine Monitoring

Sep 2025

- Reduced engine downtime by 95% by building a predictive maintenance system that analyzed 21 sensor readings from NASA Turbofan engine data using Random Forest classification to detect at-risk engines before failure, enabling proactive maintenance scheduling
- Designed React dashboard with real-time alert system and risk visualization, allowing operators to monitor 100+ engines simultaneously and automatically notify maintenance teams of critical failures
- Achieved 96% prediction accuracy with only 4.28% false alarm rate through hyperparameter tuning and feature engineering, delivering \$750K+ projected annual cost savings by preventing unplanned engine failures

Medical Recommendation System

Jan 2025 - Feb 2025

- Achieved 92% disease prediction accuracy by training and comparing SVC, Random Forest, and Gradient Boosting models on 10,000+ symptom-treatment pairs, enabling faster preliminary diagnosis for healthcare providers.
- Built a full-stack web application using Flask for RESTful API backend and responsive frontend, integrating Google Search API to deliver real-time treatment suggestions, reducing manual research time from 5+ mins to under 1 min through automated API integration

Twitter Clone | Full Stack Development | Web Application Development

May 2024- Dec 2024

- Developed a full-stack Twitter clone supporting user authentication, tweet posting, and social interactions (following/unfollowing), deployed and tested with 15+ users using React, Vite, Node.js, Express.js, MongoDB, and Cloudinary.
- Designed and implemented RESTful APIs with JWT, bcrypt, and custom middleware, applying software design best practices for scalability and security, and tested using relational database concepts to ensure reliable data handling.

ON CAMPUS INVOLVEMENT

Resident Assistant | North Central College | RA of the Year 2025

May 2024 – Present

- Increased engagement by 80% by organizing 15+ events per semester and providing mentorship to 100+ residents

Vice President | Society of Hispanic Professional Engineers (SHPE) | North Central College

May 2023 – Present

- Secured conference funding and led 10+ career development workshops for 20+ engineering students while overseeing executive board operations and coordinating semester programming

President | Computer Science Club | North Central College

May 2023 – Present

- Grew club to 25+ members by organizing 10+ technical workshops with industry professionals, leading collaborative projects, and fostering peer mentorship to increase retention and hands-on learning.

PROFESSIONAL AFFILIATIONS

Color Stack, RTC, SHPE, NSBE, AnitaB.org, Girls Who Code, ACM, Women in Technology, Codepath, Code2040